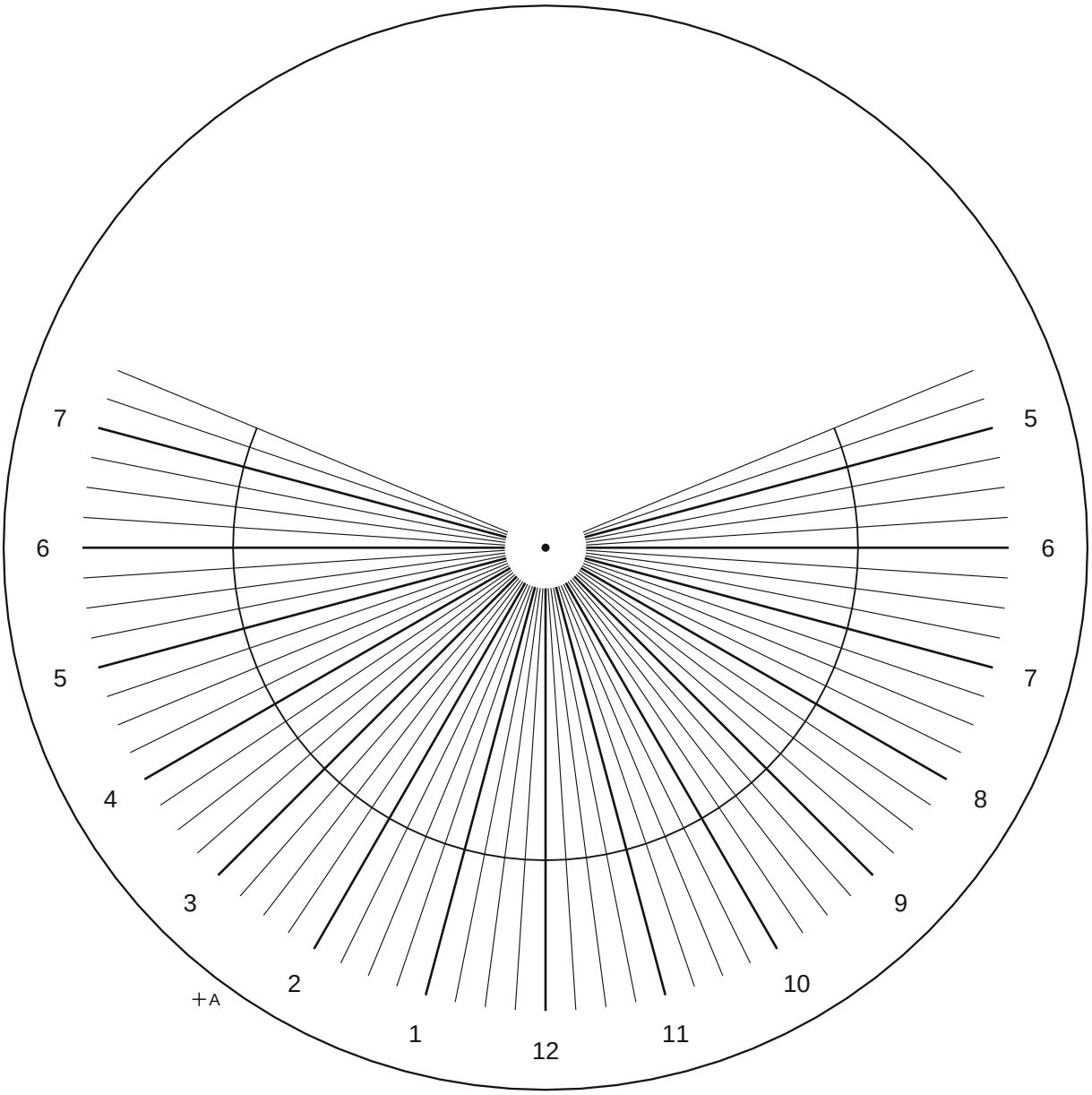
P | Pole-facing face | March equinox to September equinox | viewed from outside

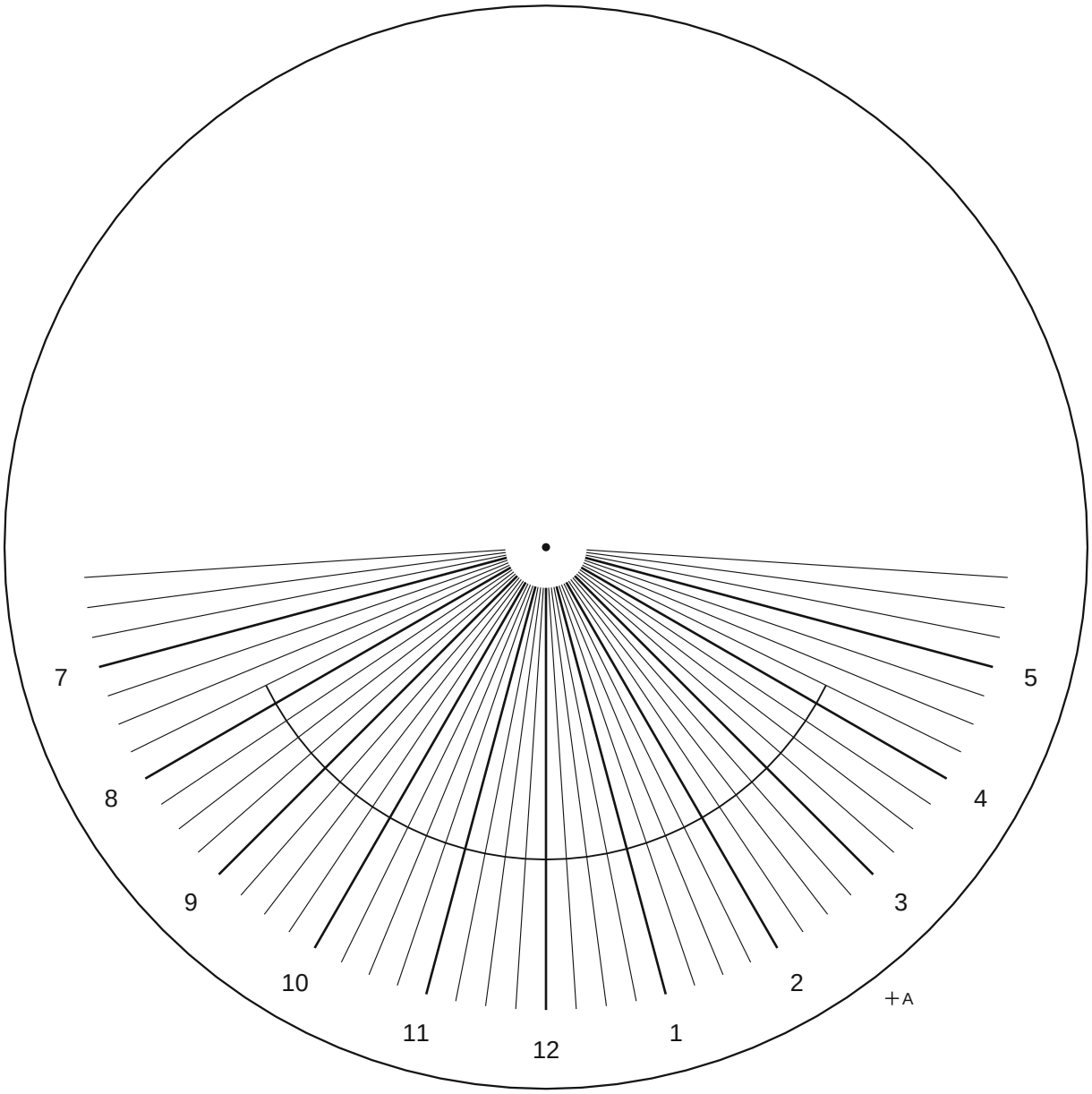


north-circle | 45 N

Controlled construction geometry, not a physical field measurement. Same fixed clock convention on both faces.

100 mm calibration - verify before cutting

O | Opposite face | September equinox to March equinox | viewed from outside



north-circle | 45 N

Controlled construction geometry, not a physical field measurement. Same fixed clock convention on both faces.

100 mm calibration - verify before cutting

Gnomon rod - length gauge

A4 | plate radius 80 mm | print at Actual size / 100%

Use a thin rod with 49.60 mm exposed on EACH side of the plate: 99.20 mm plus your backing's thickness in total. The gauge measures one exposed half from its plate surface, not the whole rod. Set the rod perpendicular through the filled foot dots.

Print both faces on separate sheets at 100%, not automatic duplex. Keep the print facing outward on each side of one rigid backing. Align the filled foot dots and target A through the backing; transfer those two marks before trimming. The outlines fit the same cut. Target A is an assembly aid outside the cut.

Away from the poles, page-up on both layouts is uphill on the tilted disc. Support the disc without blocking either face or the rod. At a geographic pole the meridian reference is arbitrary; there is no illuminated winter face during polar night.

Read the pole-facing side from March equinox to September equinox; use the opposite side in the other half of the year, when the Sun is above your horizon. At an equinox the Sun grazes both faces, so neither gives a usable reading.

For date curves, put a bead or notch 20.00 mm along EACH exposed half, measured from that side's plate surface. The cross on the gauge marks this distance; ink alone does not cast a distinct nodus shadow.

The tilt and mounting support are not cut templates. Material thickness, kerf and printer distortion are not compensated; check the calibration bars before cutting.

Tilt the disc 45.0° from horizontal, perpendicular to Earth's axis. The pole-facing side faces the north celestial pole. The rod is perpendicular to the disc. This drawing is the pole-facing face, viewed from outside.

100 mm calibration - verify before cutting